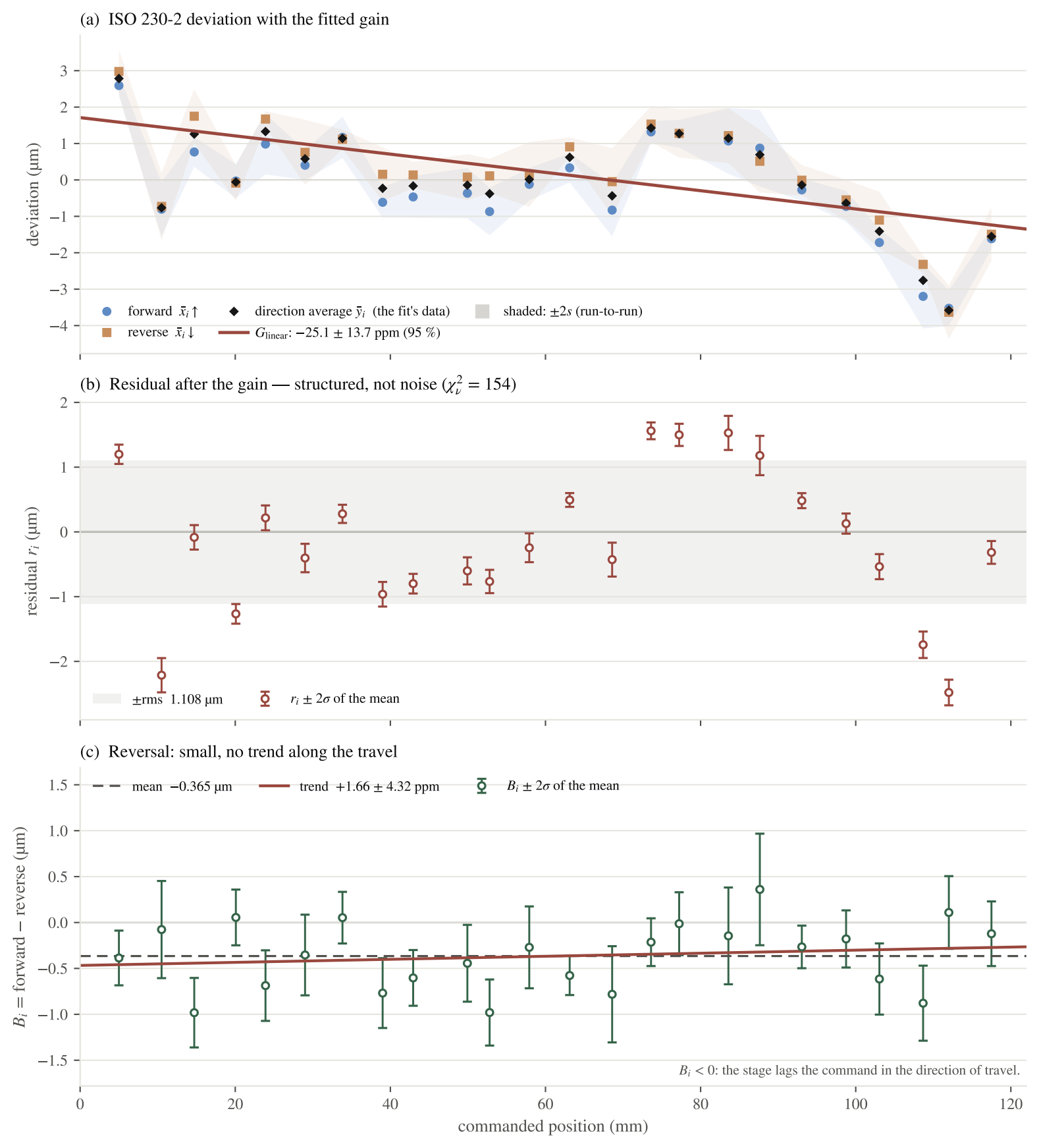
(a) ISO 230-2 deviation with the fitted gain
deviation (µm)
forward $\bar{x}_i \uparrow$
reverse $\bar{x}_i \downarrow$
direction average $\bar{y}_i$ (the fit's data)
G_{linear}: −25.1 ± 13.7 ppm (95 %)
shaded: ±2s (run-to-run)
(b) Residual after the gain — structured, not noise ($\chi^2_\nu = 154$)
residual r_i (µm)
±rms 1.108 µm
$r_i \pm 2\sigma$ of the mean
(c) Reversal: small, no trend along the travel
mean −0.365 µm
trend +1.66 ± 4.32 ppm
$B_i \pm 2\sigma$ of the mean
B_i = forward − reverse (µm)
B_i < 0: the stage lags the command in the direction of travel.
commanded position (mm)